

Keywords: distribution shift • covariate shift • concept shift
• optimal transport • generalization bounds

General Quantification of Covariate and Concept Shifts

Entropic Optimal Transport Yields the First General, Estimable Error Bound Unifying Both Shift Types

By Hongbo Chen, Li Charlie Xia (South China University of Technology)

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Generalization under distribution shift is a core unsolved challenge in modern machine learning. Existing learning bound theory handles covariate shift and concept shift in separate, narrow, idealized frameworks — and both frameworks break when the source and target distributions do not share the same support, which is the common case in deployment. This paper derives a single general bound that unifies both shift types using entropic optimal transport, is estimable from samples, and comes with an algorithm — DataShifts — that measures each component of the shift from finite data.

AT A GLANCE

Problem: Existing distribution-shift bounds are restricted to matching supports and separate shift types; neither condition holds in practice.

Why it matters: Unestimable, vacuous, or type-specific bounds cannot guide real deployment decisions under shift.

Method: Define γ^* -concept shifts via entropic OT; derive a unified bound; construct sample estimators for both components.

Result: DataShifts produces finite non-vacuous bounds on all 12 synthetic and 5 real benchmarks; 1.5–5 $\times$ tighter than prior methods where prior methods are finite.

Evidence: Formal concentration guarantees; ablations on SCM size, sample size, and cost function.

The Core Problem

A learning algorithm is trained on source data $(X_s, Y_s) \sim P_s$ and deployed on target data $(X_t, Y_t) \sim P_t$. How much does the test error on P_t exceed the training error on P_s ?

Covariate shift assumes the label mechanism is unchanged ($P_t(Y|X) = P_s(Y|X)$) but the input distribution shifts ($P_t(X) \neq P_s(X)$). Importance-weighted ERM corrects this.

Concept shift assumes the conditional itself changes: $P_t(Y|X) \neq P_s(Y|X)$. Existing bounds measure this change via f -divergences between conditionals — but these divergences are infinite when $\text{supp}(P_t) \not\subseteq \text{supp}(P_s)$, producing vacuous bounds in the mismatch setting that characterizes most real deployments.

Why Existing Bounds Fail

The authors prove a formal impossibility: any bound of the form

$$\mathcal{E}_t(h) \leq \mathcal{E}_s(h) + D_f(P_s, P_t) + \lambda$$

where D_f is any f -divergence-based concept shift measure will be infinite whenever $\text{supp}(P_t) \not\subseteq \text{supp}(P_s)$, even if the conditional distributions are in practice very close.

This is not merely a theoretical inconvenience. On standard domain adaptation benchmarks such as Camelyon17 (where hospital scanners differ) or CIFAR-10-C (where corruptions shift the input distribution globally), the target support extends beyond the source support. All prior bounds are vacuous on these benchmarks.

Entropic Optimal Transport and γ^* -Concept Shifts

The key insight is to replace f -divergence with a soft coupling via entropic optimal transport. Given a cost function $c(x, x')$ on the input space, define:

$$\gamma^* = \arg \min_{\gamma \in \Pi(P_s^X, P_t^X)} \mathbb{E}_\gamma[c(X, X')] + \varepsilon H(\gamma)$$

where $H(\gamma)$ is the entropy of the coupling and $\varepsilon > 0$ is a regularization parameter. The γ^* -**concept shift** is:

$$\Delta_\gamma = \mathbb{E}_{\gamma^*}[\text{TV}(P_s(Y|X=x), P_t(Y|X'=x'))]$$

This measures how much the label distribution changes between “matched” source and target points under the optimal transport plan. When supports agree, this recovers the classical TV-based concept shift. When supports mismatch, the transport plan couples each target point to the nearest available source point, producing a well-defined, finite quantity.

The General Error Bound

Theorem. For any hypothesis h with bounded loss $\ell \in [0, 1]$:

$$\mathcal{E}_t(h) \leq \underbrace{\mathcal{E}_s(h) + W_\varepsilon(P_s^X, P_t^X)}_{\text{covariate shift}} + \underbrace{\Delta_\gamma(P_s, P_t)}_{\gamma^*\text{-concept shift}} + O\left(\sqrt{\frac{\log |\mathcal{H}|}{n}}\right)$$

where W_ε is the entropic Wasserstein distance between input marginals. This bound:

- Applies to multiclass, regression, and stochastic labeling.
- Is finite for arbitrary source/target supports.
- Recovers known tight bounds as special cases.
- Is estimable from finite samples.

The DataShifts Algorithm

DataShifts estimates both shift components from source and target samples:

Step 1. Solve $\hat{\gamma}^* = \text{Sinkhorn}(\hat{P}_s^X, \hat{P}_t^X, c, \varepsilon)$ via Sinkhorn iterations.

Step 2. Compute $\hat{W}_\varepsilon$ as the primal OT objective.

Step 3. For each matched pair (x_i, x'_j) in $\hat{\gamma}^*$, estimate $\text{TV}(P_s(Y|x_i), P_t(Y|x'_j))$ from label empiricals.

Concentration guarantee: With probability $1 - \delta$,

$$|\hat{W}_\varepsilon - W_\varepsilon| \leq C \sqrt{\frac{\log(1/\delta)}{n}}$$

and similarly for $\hat{\Delta}_\gamma$, where C depends on the diameter of the input space.

Experimental Findings

Experiments span 12 synthetic shift scenarios and 5 real-world benchmarks (Portraits, CIFAR-10-C, Camelyon17, DomainNet, and a tabular clinical dataset):

Dataset	$\mathcal{H}\Delta\mathcal{H}$ bound	DataShifts
CIFAR-10-C (blur)	∞	0.48 (true: 0.31)
Camelyon17	2.31	0.39 (true: 0.24)
Portraits	0.87	0.27 (true: 0.18)

DataShifts is 1.5–5 $\times$ tighter where prior methods are finite, and produces finite bounds where prior methods give ∞ .

Shift Decomposition

A key practical capability of DataShifts is decomposing total shift into covariate and concept components, guiding which correction strategy to apply:

CIFAR-10-C (corruption-only): $\hat{W}_\varepsilon = 0.41$, $\hat{\Delta}_\gamma = 0.03$ — shift is almost entirely covariate; apply importance weighting.

Clinical tabular (changed deployment policy): $\hat{W}_\varepsilon = 0.07$, $\hat{\Delta}_\gamma = 0.28$ — shift is almost entirely concept; apply label calibration.

Ablations and Interpretation

Regularization ε : Smaller values tighten bounds but increase optimization difficulty; $\varepsilon = 0.1$ works well across all tested datasets.

Sample size: DataShifts requires roughly $n = 500$ target samples for reliable bounds; $n = 100$ suffices for decomposition quality.

Cost function: Euclidean cost works for image benchmarks; learned feature-space cost improves results on tabular data.

Reference

Hongbo Chen, Li Charlie Xia. **General Quantification of Covariate and Concept Shifts.** arXiv:2609.11918, ICML 2026. <https://arxiv.org/abs/2609.11918>

Keywords: topological reasoning • spatial cognition • multimodal LLM • benchmark • knot theory

MindTopo: Can Foundation Models Reason in Topological Space?

Best Model Scores 61.4% vs. 97.9% Human on a Benchmark of 11,030 Topological Intuition Instances

By Yunfei Ge, Anbang Liu, Qineng Wang, Johnalbert Garnica, Jianwen Lyu, Zihan Wang, Reuben Tan, Jianfeng Gao, Ruohan Zhang, Yining Hong, Jiajun Wu, Manling Li (Northwestern University; Microsoft Research; Stanford University; et al.)

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Spatial reasoning benchmarks for foundation models have focused almost entirely on metric properties: distances, angles, viewpoints, and shapes. Cognitive science identifies a more fundamental layer of spatial understanding — topological relations that remain invariant under continuous deformation — as the earliest spatial concepts acquired by infants. MindTopo evaluates foundation models on this layer and finds a systematic blind spot: the best multimodal LLM scores 61.4% on 11,030 topological instances against 97.9% human performance, with near-random performance on knots — the most globally structured topological property.

AT A GLANCE

Problem: Foundation models are not evaluated on topological spatial relations despite their cognitive primacy.

Why it matters: Topological reasoning underlies planning, navigation, and physical interaction; a gap here limits physical AI.

Method: 11,030 procedurally generated instances across 5 topological properties (continuity, separation, order, enclosure, knots) at reasoning and planning cognitive levels.

Result: Best model 61.4%; human 97.9%; knots near random for all 14 tested multimodal LLMs.

Evidence: 14 MLLMs + 3 video generative models evaluated; ablations with chain-of-thought, metric cue removal, difficulty scaling.

Topology vs. Metric Geometry

Topology studies properties invariant under *homeomorphisms* (continuous deformations with continuous inverses), as opposed to geometry, which studies properties

invariant under rigid motions. The classic example: a coffee cup and a donut are topologically equivalent (both have one handle/hole) but geometrically different.

Cognitive developmental science (Piaget 1956; Spelke et al. 1995) identifies topological relations as the first spatial concepts acquired by infants, preceding projective and metric relations. Despite this cognitive primacy, virtually all spatial reasoning benchmarks for foundation models measure metric or viewpoint-dependent properties.

MindTopo addresses this gap with five topological properties:

Continuity: Is a path or region connected without breaks?

Separation: Are two regions or paths disconnected?

Order: What is the cyclic ordering of points along a closed curve?

Enclosure: Does one region contain another (Jordan curve-type)?

Knots: What is the global entanglement class of a closed curve in 3D space?

The Knot Problem

Knots are the hardest property in MindTopo and the most revealing failure. A *knot* is a closed curve $\gamma : S^1 \rightarrow \mathbb{R}^3$. Two knots are equivalent (the same knot type) iff one can be deformed into the other without cutting — a relation that cannot be decomposed into any set of local features.

By the Reidemeister theorem, two knot diagrams represent equivalent knots iff one can be obtained from the other by three local moves (Reidemeister moves I, II, III). Checking equivalence requires reasoning about global path structure through an unbounded sequence of local moves. Current vision models, which process local patches and aggregate them, have no mechanism for this kind of global topological bookkeeping.

Result: All 14 MLLMs score between 24–35% on knot instances (4-choice random = 25%), confirming the problem is currently unsolvable by the tested models regardless of scale or architecture.

Benchmark Design

MindTopo contains **11,030 instances** across **13 task types** at two cognitive levels:

Reasoning: Given visual input (image or short video), identify a topological property or predict how it changes under a described deformation.

Planning: The model acts as a closed-loop agent selecting environment actions. Task completion within $T = 20$ steps is the primary metric.

Instances are generated algorithmically using a custom topology simulator with difficulty parameter $d \in [1, 5]$ controlling the number of topological operations required. All instances are procedurally generated — not scraped — ensuring no overlap with web training data.

Human baseline: 50 participants (college students with no topology training) evaluated a stratified 500-instance subset; mean accuracy 97.9%.

Technical Formulation

Let $f : X \rightarrow Y$ be a continuous map between topological spaces. MindTopo operationalizes five invariant types:

Separation: Two sets $A, B \subset X$ are separated if $\bar{A} \cap B = A \cap \bar{B} = \emptyset$.

Enclosure: Region A encloses B if $B \subset \text{Int}(A)$ in the ambient space (Jordan curve theorem generalization).

Knot equivalence: Closed curve $\gamma : S^1 \rightarrow \mathbb{R}^3$ belongs to knot class $[K]$ if it is ambient isotopic to K . MindTopo uses three-crossing and four-crossing knots as test cases.

Order invariant: Given n points on a closed curve, their cyclic order $\sigma \in S_n / \mathbb{Z}_n$ is invariant under homeomorphism.

Experimental Findings

Model	Reasoning	Planning
Human	98.2%	97.5%
Gemini 1.5 Pro (best)	68.3%	54.5%
GPT-4o	65.1%	48.9%
Claude 3.5 Sonnet	63.7%	46.2%
Avg. 14 MLLMs	52.4%	38.7%

Results by property reveal a clear hierarchy:

Property	Best Model	Human
Continuity	79.2%	98.8%
Separation	74.1%	98.5%
Order	65.8%	97.4%
Enclosure	61.3%	97.1%
Knots	29.4%	97.2%

Continuity and separation (local, visually apparent properties) are relatively easier; knots (global, non-local) are near-random.

Ablations and Interpretation

Metric cues removed: When images are topologically equivalent but metrically different (differently scaled loops), model accuracy changes by less than 2%, confirming models do not exploit metric shortcuts.

Chain-of-thought prompting: CoT improves average

accuracy by 4.1 points on reasoning and 3.8 on planning — helpful but insufficient to close the human gap.

Difficulty scaling: At difficulty $d = 5$, models average 31.2% (near random) while humans score 95.4%, indicating complex topological configurations are qualitatively out of distribution for current models.

Planning gap: All models perform better on reasoning than planning (average gap: 13.7 points), suggesting models recognize topological cues without grounding them in action-relevant planning representations.

Reference

Yunfei Ge et al. **MindTopo: Can Foundation Models Reason in Topological Space?** arXiv:2609.11900, September 2026. <https://arxiv.org/abs/2609.11900>

Keywords: long video understanding • agentic AI • edge-cloud • visual routing • multimodal LLM

Caption-once, Frames-on-Demand: Visual-Need Routing for Budget-Aware Agentic Long Video Understanding

One Offline Captioning Pass, a Dual-Track Narrative Index, and a Lightweight Router Cut Frame API Calls by 71% While Improving Accuracy

By Weitong Cai, Hang Zhang, Yukai Huang, Yiqiao Xie, Shan Gao, Jiankang Deng, Songcen Xu, Jifei Song, Zhensong Zhang (Imperial College London; Huawei Noah’s Ark Lab; et al.)

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Deploying long-video understanding on edge devices imposes two simultaneous constraints: tight compute budgets (no dense frame encoding over hours of content) and tight bandwidth budgets (no per-query frame refetch from the cloud). Prior systems sacrifice accuracy for efficiency (sparse frame methods) or efficiency for accuracy (dense MLLM methods). Caption-once, Frames-on-Demand (CFD) resolves the tradeoff: a single offline captioning pass builds a reusable dual-track narrative index, and a lightweight Visual-Need Router at query time triggers pixel retrieval only when the question requires it — 29% of the time.

AT A GLANCE

Problem: Long-video understanding on edge devices is blocked by the cost of dense frame processing and per-query frame retrieval.

Why it matters: Hours-long video QA is needed for wearables, in-car systems, and surveillance; bandwidth and compute are limited.

Method: Caption video once; build event-level story skeleton + clip-level micro-log; route each query through Visual-Need Router; retrieve frames only for perceptual sub-questions.

Result: 74.8%/60.9%/66.7% on EgoSchema/LVBench/Video-MME-Long; average 1.16 frames per query vs. 18–64 for baselines.

Evidence: Ablations on dual-track index, router design, and frame budget $R_{\max}$.

The Visual-Textual Duality

The core insight is that long-video questions separate cleanly into two types:

Structural questions (temporal order, causality, scene

transitions, narrative arc) are answered better from language memories than from dense frames. A prose description of a sequence of events conveys order and causality more compactly and reliably than 3,600 frames at 1 fps.

Perceptual questions (clothing color, on-screen text, object state, character identity) require pixels. Language captions inevitably abstract away discriminating visual details; the answer “the person is wearing a blue jacket” is useless if the question distinguishes between navy, teal, and cobalt.

Prior systems ignore this duality. Dense-frame MLLMs pay the cost of all frames for all questions. Text-only memory agents fail systematically on perceptual questions. CFD exploits the duality: build a reusable language index and retrieve frames only when pixels are genuinely needed.

The CFD Framework

Offline phase (run once at ingestion):

1. Clip segmentation. Video is divided into clips $\{c_1, \dots, c_K\}$ at scene boundaries by a local shot-detection model (no API calls).

2. Clip captioning. Each clip c_k is captioned by a compact 3B-parameter local MLLM, producing micro-log entry m_k : timestamp range, key actors/objects, described actions, verbatim on-screen text.

3. Story synthesis. A cloud MLLM writes an event-level story skeleton S (200–500 words) covering the main narrative arc, key events, and causal relationships from the full micro-log sequence.

4. Index construction. The dual-track index $\mathcal{I} = (S, \{m_k\}, \{t_k\})$ is stored locally.

Online phase (per query):

1. Story pass. Read S and identify candidate time windows $\mathcal{W}^*$ relevant to the query.

2. Micro-log pass. Read micro-logs for $\mathcal{W}^*$ and attempt a text-only answer.

3. Visual-Need Router. Classify remaining uncertainty as “perceptual” or “structural”. If structural: return text answer. If perceptual: request bounded keyframe retrieval.

4. Bounded retrieval. Request at most $R_{\max} = 4$ frames from the relevant window; integrate into the final response.

Technical Formulation

The full inference pipeline computes:

$$\hat{A} = \text{MLLM}(q, S, \{m_k : t_k \in \mathcal{W}^*\}, \{f_j : j \in \mathcal{R}(q)\})$$

where the frame retrieval set is:

$$\mathcal{R}(q) = \begin{cases} \arg \max_{|J| \leq R_{\max}} \text{Rel}(q, \{f_j\}_{j \in J}) & \phi(q, \mathcal{I}) = \text{perceptual} \\ \emptyset & \text{otherwise} \end{cases}$$

and $\phi : (q, \mathcal{I}) \rightarrow \{\text{perceptual}, \text{structural}\}$ is the Visual-Need Router.

Expected cost per query:

$$\mathbb{E}[|\mathcal{R}(q)|] = R_{\max} \cdot p_{\text{perc}} = 4 \times 0.29 = 1.16 \text{ frames}$$

vs. ~ 64 for dense-frame baselines and ~ 18 – 22 for prior agentic methods.

Visual-Need Router

The router ϕ is a 350M-parameter BERT-style encoder trained on 2,400 (question, micro-log) pairs annotated for visual necessity. It classifies a query as “perceptual” if the discriminating visual attribute cannot be recovered from the narrative text alone.

Router accuracy on 600 held-out pairs: 91.4%. Inference latency: < 1 ms on CPU — negligible vs. frame retrieval latency (200–800 ms round-trip).

Experimental Results

Method	EgoS.	LVB.	VME-L
GPT-4V-Long (dense)	72.4%	58.3%	64.1%
VideoAgent	68.9%	52.7%	59.8%
EgoVideoQA	70.3%	54.1%	61.2%
CFD (ours)	74.8%	60.9%	66.7%

EgoS. = EgoSchema; LVB. = LVBench; VME-L = Video-MME Long.

CFD improves over the best prior method by 2.4–2.8 accuracy points while using 15 – $55\times$ fewer frames per query.

Ablations and Interpretation

Without story skeleton: Accuracy drops 5.3 points on LVBench. Multi-scene temporal reasoning requires the sketch-level narrative.

Without micro-logs: Accuracy drops 3.1 points on EgoSchema. Clip-level detail is needed for fine-grained activity understanding.

Without Visual-Need Router (always retrieve): Accuracy improves by only 0.9 points but frame cost increases $3.4\times$. The router’s accuracy–efficiency tradeoff is strongly favorable.

Single-track index: Either track alone scores 4–8 points below the dual-track system, confirming the two tracks are complementary.

Reference

Weitong Cai et al. **Caption-once, Frames-on-Demand: Visual-Need Routing for Budget-Aware Agentic Long Video Understanding.** arXiv:2609.11899, EMNLP 2026. <https://arxiv.org/abs/2609.11899>

Keywords: causal discovery • foundation models • benchmark • structural causal models • memorization

CausalArena: Benchmarking Causal Discovery in the Foundation Model Era

Fixed Synthetic Benchmarks Overestimate CDFM Performance by 15–28 Points Due to Pretraining Overlap — Classical Algorithms Remain Competitive on Novel SCMs

By Zi-Rong Li, Si-Yang Liu, Tian-Zuo Wang, Han-Jia Ye (Nanjing University)

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Causal discovery has entered the foundation model era: pretrained causal discovery foundation models (CDFMs) learn to predict DAGs in-context, and LLM-based causal agents reason about structure from variable names. Both paradigms achieve impressive scores on existing benchmarks — but these benchmarks are fixed, synthetic, and drawn from the same SCM families that CDFMs were pretrained on. CausalArena introduces a unified, evolvable benchmark that controls for pretraining overlap and enables, for the first time, meaningful cross-paradigm comparisons. On genuinely novel SCMs, classical algorithms match or outperform CDFMs; on semantically rich named-variable graphs, LLM agents dominate. No paradigm is universally best.

AT A GLANCE

Problem: Fixed synthetic benchmarks inflate CDFM performance due to pretraining–test overlap; cross-paradigm comparison is impossible under inconsistent protocols.

Why it matters: Practitioners need to choose the right causal discovery tool; misleading benchmarks produce wrong choices.

Method: Procedural SCM generator producing novel graphs; common protocol with SHD, F1, and new CER metric; evolvable test set refreshed monthly.

Result: CDFMs score 8–14 SHD points worse on out-of-pretraining SCMs; classical algorithms match CDFMs on novel SCMs; LLM agents dominate on named-variable graphs.

Evidence: 11 algorithms across 3 paradigms; 12 SCM families; canary memorization test; 3 data regimes.

Causal discovery is the task of inferring a directed acyclic graph (DAG) $\mathcal{G} = (V, E)$ from observational data $\mathbf{X} \sim P_{\mathcal{G}}$. Three paradigms compete:

Classical algorithms (PC, FCI, GES, NOTEARS) operate directly on data via conditional independence tests or structural score optimization. They have no pretraining.

CDFMs (CausalFormer, DiffAN, CDFL) are pretrained on millions of synthetic SCM-generated datasets and predict DAGs in-context given a new dataset — analogous to in-context learning in language models.

LLM causal agents (GPT-4o-Causal, Claude-3.5-Causal) use semantic knowledge of variable names and domain background to reason about causal relations, augmented by statistical tests.

Each paradigm has been evaluated on different benchmarks with different graph families, mechanisms, and metrics. CDFMs report large wins over classical algorithms on their benchmark — but their benchmark uses the same SCM families they were pretrained on.

The Memorization Problem

The authors prove that CDFMs exhibit significant memorization: their performance on SCM families that appear in their pretraining corpus is 15–28 SHD points better than on structurally novel families.

Memorization gap:

$$\Delta_{\text{mem}} = \text{SHD}(M, \mathcal{G}_{\text{canonical}}) - \text{SHD}(M, \mathcal{G}_{\text{canary}})$$

where $\mathcal{G}_{\text{canary}}$ is a “canary” SCM family constructed by the authors that does not appear in any published CDFM pretraining curriculum. All three CDFMs show $\Delta_{\text{mem}} > 5$ SHD points ($p < 0.001$).

This means published CDFM benchmark numbers are inflated by pretraining memorization, not causal discovery ability.

CausalArena Design

SCM Generator. CausalArena procedurally generates novel SCMs outside any existing CDFM pretraining corpus:

- *Graph families:* Erdős–Rényi, scale-free, bipartite layered, causal motif (new).
- *Mechanisms:* Linear Gaussian, ANM, post-nonlinear, discretized multinomial.
- *Semantics:* Anonymized vs. domain-named variables.
- *Sizes:* 5–50 nodes.

Common Protocol. CausalArena enforces:

The Evaluation Crisis in Causal Discovery

- *Metrics*: SHD, F1 on directed edges, and the new Causal Effect Recovery (CER) metric.
- *Data regimes*: $n \in \{500, 5000, 50000\}$.
- *Knowledge levels*: Zero-knowledge (anonymous), partial (names), full (names + background document).

Evolvability. The test set is seeded by a public random key and refreshed monthly to prevent re-memorization. A version-controlled leaderboard tracks progress over time.

The CER Metric

SHD measures graph structure but not causal effect accuracy. CausalArena introduces **Causal Effect Recovery (CER)**:

$$\text{CER} = 1 - \frac{1}{|P|} \sum_{(i,j) \in P} \frac{|\hat{\tau}_{ij} - \tau_{ij}|}{|\tau_{ij}|}$$

where P is the set of all variable pairs, τ_{ij} is the true total causal effect of X_i on X_j (via *do*-calculus), and $\hat{\tau}_{ij}$ is estimated from the predicted DAG via structural equation fitting. CER directly measures whether the predicted graph supports correct downstream interventional reasoning.

Experimental Findings

Novel SCMs, $n = 5,000$, anonymized variables:

Algorithm	SHD ↓	F1 ↑	CER ↑
GES	7.9	0.64	0.74
PC	8.4	0.61	0.71
CausalFormer	9.8	0.57	0.65
DiffAN	10.3	0.54	0.63
GPT-4o-Causal	18.7	0.31	0.41

Classical algorithms outperform CDFMs on novel, anonymized SCMs — reversing the apparent superiority CDFMs show on fixed benchmarks.

Named variables, $n = 5,000$:

Algorithm	SHD ↓	F1 ↑	CER ↑
GES	8.1	0.63	0.73
CausalFormer	10.1	0.56	0.64
GPT-4o-Causal	6.2	0.74	0.82

LLM agents dominate when variable names carry domain-specific semantics.

Ablations and Interpretation

Sample size: CDFMs show flat performance across $n \in \{500, 5000, 50000\}$ — they rely on learned structure priors. Classical algorithms improve monotonically with n .

Mechanism complexity: Post-nonlinear mechanisms

are hardest for all methods; CDFMs trained on linear Gaussian data show a 12-point SHD gap on nonlinear instances.

Graph size: All algorithms degrade with more nodes; classical algorithms degrade more gracefully at $n > 20$ nodes than CDFMs.

Semantic grounding: LLM agents degrade sharply on anonymized variables (−12.5 SHD points), showing their advantage is entirely domain-knowledge driven.

Reference

Zi-Rong Li, Si-Yang Liu, Tian-Zuo Wang, Han-Jia Ye. **CausalArena: Benchmarking Causal Discovery in the Foundation Model Era.** arXiv:2609.11897, September 2026. <https://arxiv.org/abs/2609.11897>

Keywords: KV cache quantization • omni-modal LLM • multimodal inference • 2-bit quantization • efficient serving

OmniKVQuant: KV Cache Quantization for Omni-LLMs

Temporal Key Drift and Heterogeneous Value Geometry Break Existing 2-Bit Methods on Multimodal Caches — Per-Window and Per-Modality Fixes Achieve Near-Lossless Quality

By Suho Yoo, Hyunjong Ok, Jongmin Choi, Jihoo Jung, Joon Son Chung (KAIST; POSTECH)

arXiv:2609.11582

Omni-modal large language models — models that jointly process audio, video, and text — are increasingly used in real-time deployments. Their KV cache grows proportionally to all modalities simultaneously: a 7B omni-model processing 10 minutes of audio-video accumulates ~ 30 GB of FP16 KV cache. 2-bit quantization would reduce this to 4 GB, enabling consumer deployment. But applying existing rotation-based quantization methods (designed for text-only LLMs) to multimodal caches cuts performance by 20 points or more. OmniKVQuant diagnoses why — two failure modes specific to multimodal caches — and fixes them with targeted, training-free interventions.

AT A GLANCE

Problem: Rotation-based KV cache quantization fails on omni-modal caches; no training-free solution exists.

Why it matters: Omni-LLMs for real-time audio-video require radical memory reduction; 2-bit is the practical target.

Method: Per-window key quantization (for temporal key drift); per-modality value rotation (for heterogeneous value geometry); fused Triton decode kernel.

Result: 2-bit KV cache on Qwen2.5-Omni and Qwen3-Omni with only 2.3-point degradation vs. 19–24 points for prior methods.

Evidence: 7 audio-visual benchmarks; ablations on window size, modality rotation, and kernel overhead.

KV Cache Memory in Omni-LLMs

The KV cache size scales as:

$$M_{KV} = 2 \cdot n_l \cdot n_h \cdot L \cdot d_h \cdot \text{prec}$$

For a 7B omni-model (32 layers, 32 heads, $d_h = 128$) processing 10 minutes of audio-video at 25 fps ($L \approx 15,000$), this is ≈ 30 GB at FP16 — far beyond consumer GPU memory.

Existing KV cache quantization for text-only LLMs uses rotation-based methods: apply an orthogonal matrix R to key/value vectors before quantization to eliminate outliers:

$$\tilde{V} = RV, \quad \hat{V} = Q_2(\tilde{V})$$

where Q_2 is 2-bit rounding. R is calibrated once on text calibration data. On text-only LLMs, this reduces memory to ≈ 4 GB with <1 point degradation. On omni-modal caches, applying the same R causes 19–24 point degradation.

Failure Mode 1: Temporal Key Drift

In a text-only LLM, the statistical distribution of key vectors is roughly stationary over the sequence. In an omni-modal model processing audio:

$$\mathbf{K}_t = W_K \cdot \mathbf{x}_t^{\text{audio}}$$

where $\mathbf{x}_t^{\text{audio}}$ is the audio embedding at time t . Audio embeddings have strong temporal autocorrelation — consecutive frames are highly similar — so the key distribution drifts systematically over time:

$$\|\mathbf{K}_t - \mathbf{K}_{t-1}\|_2 \ll \|\mathbf{K}_t\|_2$$

A quantization range $[r_{\min}, r_{\max}]$ calibrated at initialization clips new key values that drift outside this range, producing growing quantization error over longer audio streams.

Fix: per-window key quantization. Apply separate quantization ranges to each temporal window of size w :

$$r_{\min}^{(t)} = \min_{t' \in [t-w, t]} K_{j, t'}^{\min}, \quad r_{\max}^{(t)} = \max_{t' \in [t-w, t]} K_{j, t'}^{\max}$$

This adapts the quantization range locally, tracking drift without requiring global statistics. Window size $w = 64$ works well across all tested models.

Failure Mode 2: Heterogeneous Value Geometry

The optimal rotation R_m^* for modality m is the PCA rotation of the value covariance matrix Σ_m :

$$R_m^* \approx U_m^\top, \quad \Sigma_m = U_m \Lambda_m U_m^\top$$

These matrices are modality-specific:

- *Text:* Near-isotropic; ~ 30 directions with $>1\%$ variance.
- *Audio:* Concentrated; 90% variance in 5 low-frequency pitch/timbre directions.
- *Video:* High-rank anisotropic; large dynamic range in spatial-frequency directions.

Applying R_{text}^* to audio values leaves 85% of variance concentrated in 5 directions, producing massive outliers in the rotated space that 2-bit quantization cannot faithfully represent.

Fix: per-modality value rotation. Calibrate and store separate rotations $\{R_m^*\}$ per modality. At inference, detect the current modality from the token type embedding and apply the appropriate rotation:

$$\tilde{\mathbf{V}} = R_m^* \mathbf{V}, \quad \hat{\mathbf{V}} = Q_2(\tilde{\mathbf{V}})$$

Fused Triton Decode Kernel

Standard 2-bit implementations dequantize the cache to FP16 before attention, reconstructing the full dense cache and negating half the memory savings. OmniKVQuant provides a fused Triton kernel that dequantizes on the fly during attention computation, reading 2-bit integers per block and expanding to FP16 without storing the FP16 cache. Memory footprint remains at 2 bits throughout.

2-bit symmetric quantization:

$$\hat{v}_j = s \cdot \text{round}\left(\frac{v_j}{s}\right), \quad \text{round}(\cdot) \in \{-2, -1, 0, 1\}, \quad s = \frac{\max_j |v_j|}{2}$$

Experimental Results

Tested on Qwen2.5-Omni-7B and Qwen3-Omni-7B across 7 audio-visual benchmarks (LibriSpeech, MMAU, Video-MME, ActivityNet-QA, MSVD-QA, OmniBench, Evo-OMNIBench):

Method	Bits	Avg Score	Δ
Full FP16	16	72.4	—
TurboQuant	2	48.3	−24.1
KIVI	2	52.7	−19.7
OmniKVQuant	2	70.1	−2.3
OmniKVQuant	4	72.0	−0.4

OmniKVQuant reduces 2-bit degradation from 19–24 points (prior methods) to 2.3 points, and achieves near-lossless 4-bit quantization (0.4 points).

Ablations and Interpretation

Without per-window key quantization: Degradation grows from 2.3 to 11.4 points; confirms temporal key drift is the dominant failure mode.

Without per-modality value rotation: Degradation grows from 2.3 to 8.9 points; audio benchmarks are disproportionately affected (−18.3 on MMAU).

Kernel overhead: The fused Triton kernel adds 4% latency vs. FP16 attention while delivering $7.8\times$ memory reduction for 2-bit vs. FP16 caches.

Window size: $w = 64$ works well; smaller w slightly improves accuracy at the cost of more quantization parameter updates.

Reference

Suho Yoo, Hyunjong Ok, Jongmin Choi, Jihoo Jung, Joon Son Chung. **OmniKVQuant: KV Cache Quantization for Omni-LLMs.** arXiv:2609.11582, September 2026. <https://arxiv.org/abs/2609.11582>